Fiscal Year:	FY 2005	**Task Last Updated:**	FY 04/27/2005
PI Name:	Megaridis, Constantine M. Ph.D.		
Project Title:	Molten-Metal Droplet Deposition on a Moving Substrate in Microgravity: Aiding the Development of Novel Technologies for Microelectronic Assembly		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Chicago	**State:**	IL
Zip Code:	60607-7022	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	07/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Duval, W.	**Contact Phone:**	216-433-5023
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Flight Program:			
Flight Assignment:	note: extended 1 year		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Poulikakos, Dimos (Swiss Federal Inst. of Technology (ETH Zurich)) Nayagam, Vedha (NCMR)		
Grant/Contract No.:	NAG3-2456		
Performance Goal No.:	1.23		
Performance Goal Text:	Pursue ground-breaking basic research in fluid physics and transport phenomena to provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.		

Task Description:	The study consists of a theoretical and an experimental component. The theoretical work uses computer models based on finite element techniques. These models require, as a boundary condition, the relationship between contact angle and contact line speed in droplet impact configurations. The experimental work analyzes droplet impacts to extract information on the dynamics of the contact line. The objective of this program is to investigate the fluid dynamics and simultaneous solidification of molten solder droplets impacting on a flat surface. The problem is directly relevant to the printing of microscopic solder droplets during surface mounting of microelectronic devices. The study is pursued in collaboration with an industrial partner, MicroFab Technologies, and seeks to identify the degree of influence of the dominant process parameters on specific aspects of solder droplet dispensing. The data produced is analyzed to define the domains of operational parameters that result in optimal and consistent production of solder bumps, as needed for specific industrial applications. The technology of solder microdroplet dispensing has shown good promise in microelectronic packaging and assembly. Therefore, the development of a good understanding of the pertinent fluid dynamics and solidification phenomena is essential for the successful commercial implementation of the technique. Conducting experiments in microgravity eliminates the unwanted influence of gravity and makes the experimental investigation of large droplet dispension directly relevant to the actual application, where solder droplets are approximately four thousands of an inch in diameter. This program advances the state of knowledge and provides a science base for the fundamental transport phenomena occurring in the solder microdroplet dispensing technology. The study also seeks to produce predictive tools to be used for the successful development of this innovative technique in electronic component manufacturing or other relevant technologies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Contact angle dynamics during spreading/recoiling of mm-sized water droplets impacting orthogonally on various surfaces with We = O(0.1)-O(10), Ca = O(0.001)- O(0.01), Re = O(100) - O(1000), Oh = O(0.001) and Bo = O(0.1) has been studied in detail. In this impact regime, inertial, viscous, and capillary phenomena act in unison to influence contact angle dynamics. The wetting properties of the target surfaces range from wettable to non-wettable. The experiments feature accelerating and decelerating wetting lines, capillary surface waves in the early impact stages, contact angle hysteresis, and droplet rebound under non-wetting conditions. The work provided insight into the dynamic behavior of the apparent (macroscopic) contact angle and its dependence on contact line velocity at various degrees of surface wetting for droplets impacting with low to moderate Weber numbers. The results revealed that surface wettability have a critical influence on contact angle dynamic behavior. The contact angle vs. contact line speed correlations of Hoffman and Cox were employed to fit the present measurements, as well as the data reported by other investigators. These comparisons showed mixed success in the impact driven spreading regime under study. The modified molecular-kinetic theory of wetting by Blake fit the data well for impacts on wettable and partially wettable surfaces, yielding physically acceptable molecular parameters describing the wetting line spreading mechanism. The results offer guidance for numerical or analytical studies, which require careful attention to the implementation of boundary conditions, including the need to specify the dependence of contact angle on contact line speed.
Bibliography Type:	Description: (Last Updated: 06/10/2005)
Articles in Peer-reviewed Journals	Megaridis, C.M.; Boomsma, K.; Bayer, I.S. "Partial Rebound of Molten-Metal Droplets Impacting on Solid Substrate" AIChE Journal 50, 1356-1363 , Jul-2004
Papers from Meeting Proceedings	Bayer, I. S.; Megaridis, C. M. "Contact Angle Dynamics of Droplets Impacting on Flat Substrates" XXI International Congress of Theoretical and Applied Mechanics, Warsaw, Poland Aug-2004
Presentation	Bayer, I. S.; Megaridis, C. M. "Contact Angle Dynamics of Droplets Impacting on Flat Substrates" Fourth International Symposium on Contact Angle Wettability and Adhesion, Philadelphia, PA Jun-2004